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Ques 1: Write Java Program to Check Leap Year

Code:

```java
import java.util.Scanner;
public class ques1 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter a year: ");
    int year = sc.nextInt();

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
      System.out.println(year + " is a Leap Year.");
    } else {
      System.out.println(year + " is not a Leap Year.");
    }
    sc.close();
  }
}
```

Output:

```
PS C:\SEM 5\Enterprise Web Application\Lab
ques1.java } ; if ($?) { java ques1 }
Enter a year: 2030
2030 is not a Leap Year.
```

Ques 2 : Write Java Program to Check Whether a Character is Alphabet or Not

Code:

```java
import java.util.Scanner;
public class ques2 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter a character: ");
    char ch = sc.next().charAt(0);

    if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z')) {
      System.out.println(ch + " is an alphabet.");
    } else {
      System.out.println(ch + " is not an alphabet.");
    }
    sc.close();
  }
}
```

Output:

```
● ques2.java } ; if ($?) { java ques2 }
  Enter a character: s
  s is an alphabet.
```

Ques3 : Write Java Program to Find Factorial of a Number

Code :

```
import java.util.Scanner;
public class ques3 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter a number: ");
    int n = sc.nextInt();

    long fact = 1;
    for (int i = 1; i <= n; i++) {
      fact *= i;
    }

    System.out.println("Factorial of " + n + " = " + fact);
    sc.close();
  }
}
```

Output :

```
($?) { javac ques3.java } ; if
Enter a number: 5
Factorial of 5 = 120
```

Ques 4 : Write Java Program to Display Fibonacci Series

Code :

```
import java.util.Scanner;
public class ques4 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter number of terms: ");
    int n = sc.nextInt();
    int t1 = 0, t2 = 1;
    System.out.print("Fibonacci Series: ");
    for (int i = 1; i <= n; i++) {
      System.out.print(t1 + " ");
      int sum = t1 + t2;
      t1 = t2;
      t2 = sum;
    }
    sc.close(); } }
```

Output :

```
Enter number of terms: 8
Fibonacci Series: 0 1 1 2 3 5 8 13
```

Ques 5 : Write Java Program to Find GCD of two Numbers

Code :

```java
import java.util.Scanner;
public class ques5 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter first number: ");
    int a = sc.nextInt();
    System.out.print("Enter second number: ");
    int b = sc.nextInt();

    int x = a, y = b;
    while (y != 0) {
      int temp = y;
      y = x % y;
      x = temp;
    }

    System.out.println("GCD of " + a + " and " + b + " = " + x);
    sc.close();
  }
}
```

Output :

```
Enter first number: 196
Enter second number: 116
GCD of 196 and 116 = 4
```

Ques 6 : Write Java Program to Find LCM of two Numbers

Code :

```
import java.util.Scanner;
public class ques6 {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter first number: ");
      int a = sc.nextInt();
      System.out.print("Enter second number: ");
      int b = sc.nextInt();

      int gcd = findGCD(a, b);
      int lcm = (a * b) / gcd;

      System.out.println("LCM of " + a + " and " + b + " = " + lcm);
      sc.close();
   }

   static int findGCD(int a, int b) {
      while (b != 0) {
         int temp = b;
         b = a % b;
         a = temp;
      }
      return a;
   }
}
```

Output :

```
Enter first number: 17
Enter second number: 14
LCM of 17 and 14 = 238
```

Ques 7 : Write Java Program to Count Number of Digits in an Integer

Code :

```
import java.util.Scanner;
public class ques7 {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter an integer: ");
      int n = sc.nextInt();

      int num = Math.abs(n);
      int count = 0;

      if (num == 0) {
         count = 1;
      } else {
         while (num != 0) {
            num /= 10;
```

```
            count++;
          }
        }

        System.out.println("Number of digits in " + n + " = " + count);
        sc.close();
    }
}
```

Output :

```
Enter an integer: 144
Number of digits in 144 = 3
```

Ques 8 : Write Java Program to Reverse a Number

Code :

```
import java.util.Scanner;
public class ques8 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int num = Math.abs(n);
        int reversed = 0;

        while (num != 0) {
            int digit = num % 10;
            reversed = reversed * 10 + digit;
            num /= 10;
        }

        if (n < 0) {
            reversed = -reversed;
        }

        System.out.println("Reversed number = " + reversed);
        sc.close();
    }
}
```

Output :

```
Enter a number: 135
Reversed number = 531
```

Ques 9 : Write Java Program to Calculate the Power of a Number

Code :

```
import java.util.Scanner;
```

```java
public class ques9 {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter base: ");
      double base = sc.nextDouble();
      System.out.print("Enter exponent (non-negative integer): ");
      int exp = sc.nextInt();

      double result = 1;
      for (int i = 0; i < exp; i++) {
         result *= base;
      }

      System.out.println(base + " ^ " + exp + " = " + result);
      sc.close();
   }
}
```

Output :

```
Enter base: 6
Enter exponent (non-negative integer): 3
6.0 ^ 3 = 216.0
```

Ques 10 : Write Java Program to Check Palindrome

Code :

```java
import java.util.Scanner;
public class ques10 {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter a number: ");
      int n = sc.nextInt();

      int num = n, reversed = 0;
      while (num != 0) {
         int digit = num % 10;
         reversed = reversed * 10 + digit;
         num /= 10;
      }

      if (n == reversed) {
         System.out.println(n + " is a Palindrome.");
      } else {
         System.out.println(n + " is not a Palindrome.");
      }
      sc.close();
   }
}
```

Output :

```
Enter a number: 14541
14541 is a Palindrome.
```